

SANITIZABLE CASE FOR VR EQUIPMENT

Wipe-clean case for VR in the clinic

Hard cases hold VR headsets in foam, and foam can't be disinfected. This project keeps the rugged case and replaces the foam with one seamless, vacuum-formed tray that a clinician can wipe clean in a single pass.

Developed and tested with clinical users and put into use by doctors at the University of Kentucky hospital. The headset, controllers, charger and cable each sit in their own shaped recess, so cleaning is one wipe and a missing item shows up as an empty shape.

USE	Clinicians at the University of Kentucky hospital
MEDIUM	Vacuum-formed PETG / HIPS tray, about 1.5 mm, over a CNC-milled buck, in a Pelican 1550
KIT	Meta Quest 3, Touch Plus controllers, charger, USB-C cable
RECORD	UK Invention Report 2770, 2023

THE PROBLEM

Foam is the part you can't clean.

A VR headset in a hospital moves from room to room and rests against people's faces, so it gets wiped with disinfectant between uses. The headset can be wiped. The case it travels in usually can't.

Pick-and-pluck foam soaks up liquid, holds debris in its pores and cut faces, and breaks down under repeated disinfectant.

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Pelican 1550 with the foam removed and the formed insert fitted, shown with the charger and cable in their pockets. Render. [Watch the case-opening animation: siatohi80.github.io/vr-case-study](#)

APPROACH

Keep the case, replace only the foam.

A standard Pelican case is already sealed, crushproof and familiar to hospital staff. The only part that fails the cleaning test is the foam, so a single formed sheet takes its place, with a recess for every item.

STOCK FOAM

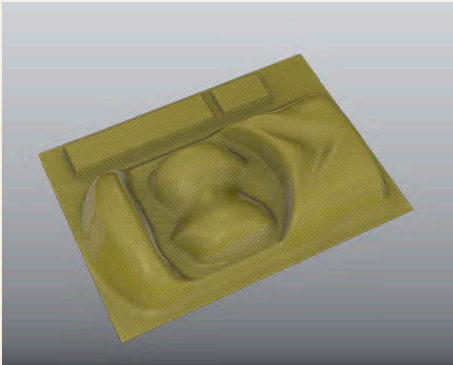
- Porous, absorbs spills and cleaner
- Cut edges and gaps trap debris
- Loose parts end up anywhere

FORMED TRAY

- One continuous, non-porous surface
- Rounded bays, no seams to trap debris
- Each item has its own shaped place

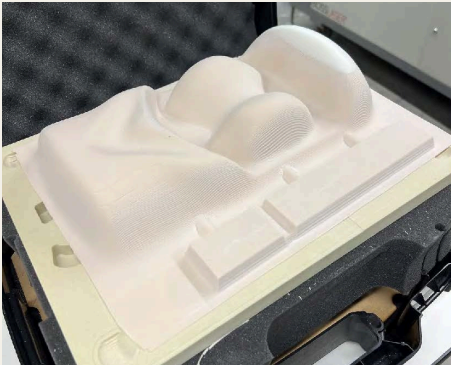
PROCESS

From surface model to fitted tray



STEP 1
Shape the form

One surface in Rhino, following the headset's strap and lens side, the dome of its front shell, and the controllers.



STEP 2
Mill the buck

The form was CNC-milled as a positive mold and test-fitted in the case. Raised blocks become the charger and cable pockets.



STEP 3
Vacuum-form the tray

A heated PETG or HIPS sheet, about 1.5 mm thick, is pulled over the buck into a thin, rigid shell with smooth radii everywhere a cloth needs to reach.



STEP 4
Fit and hand over

The tray drops into the case in place of the foam; the lid foam stays. The finished case went into use at the UK hospital.



The insert rebuilt for the Pelican 1550 and Meta Quest 3: the headset and controller bays keep the tested shape, and charger and cable pockets with finger notches line the front edge. Render.

FINAL PRODUCT · PELICAN 1550 INSERT

ITEM	VALUE
Case interior	18.63 × 14.18 × 7.72 in
Tray	18.50 × 14.05 in, 1.5 mm
Headset bay	13.9 × 7.4 in, 1.9–2.6 in deep
Charger pocket	3.10 × 2.40 × 1.55 in
Cable pocket	6.60 × 1.90 × 1.05 in
Support	Flange at the rim, formed skirt to floor

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“Sanitizable Case for VR Equipment”

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